

Trust-Calibrated Multi-Agent Scientific Deliberation for Mitigating Sycophantic Consensus in LLM Reasoning

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Abstract

Large language models have done surprisingly well at answering many scientific questions. However, large language models also continue to "hallucinate" and reason incorrectly. One potential solution to these problems is multi-agent debate. Although a correct minority may exist, a confident but incorrect majority can drag this correct minority agent down and lead the group to select the wrong answer. This is known as the "sycophantic consensus." Currently, there is no mechanism within any debate system to check an individual agent's assertions against outside scientific evidence before allowing them to gain influence in the debate. Instead of using a simple vote-weighted approach, we intend to assign a trust score to every participating agent based on how much scientific evidence supports their responses. Every time agents respond during a round, the system breaks down their responses into small units of factually verifiable statements ("claims"). After breaking down the response, the system locates all relevant scientific literature that could support or contradict each claim. Because unsupported claims eventually lose influence and because the trust scores are kept in bounds, the probability increases that a correct minority will survive long enough to ultimately win. Our proposed framework produced fewer hallucinated answers than competing frameworks because all surviving claims are linked back to their originating passages of scientific evidence. To test this hypothesis, we compare our proposed framework to majority voting and other baseline systems. Once all models have finished responding to a given question, the system returns both the selected model and supporting citations for that model. The performance of our proposed framework is evaluated via four metrics: answer accuracy, consensus collapse, minority preservation, and evidence calibration.

Keywords: Multi-agent debate, Large language models, Sycophancy, Trust calibration, Retrieval-augmented generation, Scientific question answering

1. Introduction

Large language models can answer scientific questions at a level that makes them useful for tutoring and decision support. A model can still be confidently wrong,

and it can state a wrong answer in fluent language that hides the error. Multi-agent debate reduces this risk because three agents answer the same question and review each other’s reasoning over rounds [1]. Chain-of-thought prompting lets each agent expose the steps behind its answer, so peers can inspect the reasoning [2]. The weak point sits at the end of the debate. Most systems combine the answers by majority vote, where every agent counts the same and the most frequent answer wins [3, 4]. A confident and wrong majority can then push a correct minority agent to drop its answer, and the vote records the wrong result as agreement. This failure is called sycophantic consensus [5, 6]. No current debate system checks an agent’s claims against outside scientific evidence before that agent earns influence. This study builds a trust-calibrated multi-agent deliberation framework to close that gap. Each agent’s answer is split into small factual claims. The claims are checked against retrieved scientific literature, and every agent holds a bounded trust score that rises with supporting evidence and falls with contradicting evidence. The final answer comes from trust-weighted aggregation instead of a head count. The method is tested on scientific reasoning datasets such as GPQA and MMLU-Pro, and it needs no fine-tuning because all models run as served inference.

Du et al. show that debate improves factuality and reasoning over a single pass, but their agents are treated equally and their claims are never checked against external sources [1]. Pitre et al. report that the correct answer is present in the discussion yet ignored in more than 20 percent of wrong-answer cases, and their prompt-rewriting fix still leaves the final decision on self-reported confidence [5]. The uncertainty route of Yoffe et al. reweights agent influence during the debate, and their own ground-truth experiment shows the gain is capped by the quality of that self-reported signal [7]. Token cost is the target of Fan et al., who decide when a debate is worth running, yet majority voting remains once the debate starts [8]. Wang et al. combine many models through static aggregation and score well on open benchmarks, with no per-agent trust and no evidence check [9]. He et al. detect a suppressed correct minority after the debate and overturn the majority, which comes too late to protect the minority while it argues [6].

Across these systems, influence is set by a vote, a self-reported score, a prompt rewrite, or a post-hoc flip, and none of them verifies an agent’s claims against evidence while the debate is running. Self-reported confidence is a weak basis because training that makes models agreeable also makes them more sycophantic and less accurate [10], and alignment training has been shown to increase agreement with wrong user positions [11]. Theory also predicts that correlated agents drift to a wrong majority unless an outside correction enters the loop [12], and multi-agent systems fail in recurring ways that diagnosis alone does not fix [13]. This study resolves the unsolved part by tying each agent’s influence to external evidence during the debate. Claims are checked against retrieved passages from scientific sources [14, 15], a step in line with work that grounds correction in external tools [16]. The trust score is bounded, so no agent is silenced or made all-powerful, and the final decision uses those weights. Scientific question sets are a suitable testbed because they carry verifiable answers and published literature that retrieval can

reach, and GPQA and MMLU-Pro allow comparison against established baselines. The expected outcome is a reproducible framework that lowers consensus collapse, preserves a correct minority more often, keeps answer accuracy stable, and reports a calibrated trust signal.

2. Related Work

3. Proposed Framework

4. Implementation

5. Evaluation

6. Discussion

7. Conclusion and Future Work

References

- [1] Y. Du, S. Li, A. Torralba, J. B. Tenenbaum, I. Mordatch, Improving factuality and reasoning in language models through multiagent debate, in: Proceedings of the 41st International Conference on Machine Learning, Vol. 235, 2024, pp. 11733–11763.
URL <https://proceedings.mlr.press/v235/du24e.html>
- [2] J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. H. Chi, Q. V. Le, D. Zhou, Chain-of-thought prompting elicits reasoning in large language models, in: Advances in Neural Information Processing Systems, NIPS '22, Curran Associates Inc., Red Hook, NY, USA, 2022.
URL <https://arxiv.org/abs/2201.11903>
- [3] Q. Wang, Z. Wang, Y. Su, H. Tong, Y. Song, Rethinking the bounds of LLM reasoning: Are multi-agent discussions the key?, in: L.-W. Ku, A. Martins, V. Srikumar (Eds.), Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics, Bangkok, Thailand, 2024, pp. 6106–6131. doi: 10.18653/v1/2024.acl-long.331.
URL <https://aclanthology.org/2024.acl-long.331/>
- [4] H. K. Choi, J. Zhu, S. Li, Debate or vote: Which yields better decisions in multi-agent large language models?, in: Advances in Neural Information Processing Systems, 2025.
URL <https://openreview.net/forum?id=iUjGNJzrF1>
- [5] P. Pitre, N. Ramakrishnan, X. Wang, CONSENSAGENT: Towards efficient and effective consensus in multi-agent LLM interactions through sycophancy mitigation, in: Findings of the Association for Computational Linguistics: ACL 2025, Association for Computational Linguistics, Vienna, Austria, 2025, pp. 22112–22133. doi: 10.18653/v1/2025.findings-acl.1141.
URL <https://aclanthology.org/2025.findings-acl.1141/>
- [6] C. He, Z. Chen, Z. Yang, S. Qiao, M. Ju, J. Liu, D. Wen, G. Liu, Minority sentinel: When to overturn majority voting in multi-agent LLM debates, in: Proceedings of the AgentSearch Workshop at SIGIR 2026, Melbourne, Australia, 2026, arXiv:2606.29270. doi: 10.48550/arXiv.2606.29270.
URL <https://arxiv.org/abs/2606.29270>
- [7] L. Yoffe, A. Amayuelas, W. Y. Wang, DebUnc: Improving large language model agent communication with uncertainty metrics, in: Findings of the Association for Computational Linguistics: EMNLP 2025, Association for Computational Linguistics, Suzhou, China, 2025, pp. 23299–23315. doi: 10.18653/v1/2025.findings-emnlp.1265.
URL <https://aclanthology.org/2025.findings-emnlp.1265/>

- [8] W. Fan, J. Yoon, B. Ji, iMAD: Intelligent multi-agent debate for efficient and accurate LLM inference, in: Proceedings of the Fortieth AAAI Conference on Artificial Intelligence, 2026, pp. 29403–29411. doi:10.1609/aaai.v40i35.40181.
- [9] J. Wang, J. Wang, B. Athiwaratkun, C. Zhang, J. Y. Zou, Mixture-of-agents enhances large language model capabilities, in: Y. Yue, A. Garg, N. Peng, F. Sha, R. Yu (Eds.), International Conference on Learning Representations, Vol. 2025, 2025, pp. 33944–33963.
URL https://proceedings.iclr.cc/paper_files/paper/2025/file/5434be94e82c54327bb9dcdf7fca52b6-Paper-Conference.pdf
- [10] L. Ibrahim, F. S. Hafner, L. Rocher, Training language models to be warm can reduce accuracy and increase sycophancy, *Nature* 652 (2026) 1159–1165. doi:10.1038/s41586-026-10410-0.
- [11] C.-H. Chen, H.-H. Huang, H.-H. Chen, Self-augmented preference alignment for sycophancy reduction in LLMs, in: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, Suzhou, China, 2025, pp. 12379–12391. doi:10.18653/v1/2025.emnlp-main.625.
URL <https://aclanthology.org/2025.emnlp-main.625/>
- [12] A. Estornell, Y. Liu, Multi-LLM debate: Framework, principals, and interventions, in: Advances in Neural Information Processing Systems, Vol. 38, 2024.
URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/32e07a110c6c6acf1afbf2bf82b614ad-Abstract-Conference.html
- [13] M. Cemri, M. Z. Pan, S. Yang, L. A. Agrawal, B. Chopra, R. Tiwari, K. Keutzer, A. Parameswaran, D. Klein, K. Ramchandran, M. Zaharia, J. E. Gonzalez, I. Stoica, Why do multi-agent LLM systems fail?, in: Advances in Neural Information Processing Systems, 2025, datasets and Benchmarks Track.
URL <https://arxiv.org/abs/2503.13657>
- [14] V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, W.-t. Yih, Dense passage retrieval for open-domain question answering, in: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), Association for Computational Linguistics, Online, 2020, pp. 6769–6781. doi:10.18653/v1/2020.emnlp-main.550.
URL <https://aclanthology.org/2020.emnlp-main.550/>
- [15] P. Kushwaha, S. A. Madhavan, Agent-based contradiction detection and resolution for multi-source retrieval augmented generation systems, *International Journal of Scientific Research in Engineering and Management*

- (IJSREM) 10 (7) (Jul. 2026).
 URL <https://ijsrem.com/uploads/articles/1783285322570-ecd0e35bd42a-Research-paper-2.pdf>
- [16] Z. Gou, Z. Shao, Y. Gong, Y. Shen, Y. Yang, N. Duan, W. Chen, CRITIC: Large language models can self-correct with tool-interactive critiquing, in: The Twelfth International Conference on Learning Representations, 2024.
 URL <https://arxiv.org/abs/2305.11738>
- [17] X. Y. Lee, S. Akatsuka, L. Vidyaratne, A. Kumar, A. Farahat, C. Gupta, Reliable decision-making for multi-agent LLM systems, in: Workshop on Advancing LLM-Based Multi-Agent Collaboration at AAAI 2025, 2025.
 URL https://multiagents.org/2025_artifacts/reliable_decision_making_for_multi_agent_llm_systems.pdf
- [18] E. Cau, V. Pansanella, D. Pedreschi, et al., Selective agreement, not sycophancy: investigating opinion dynamics in LLM interactions, EPJ Data Science 14 (2025) 59. doi:10.1140/epjds/s13688-025-00579-1.
- [19] Y. Cui, H. Fu, H. Zhang, L. Wang, C. Zuo, FREE-MAD: Consensus-free multi-agent debate, in: Findings of the Association for Computational Linguistics: ACL 2026, 2026.
 URL <https://aclanthology.org/2026.findings-acl.1600.pdf>
- [20] H. Zhou, H. Geng, X. Xue, L. Kang, Y. Qin, Z. Wang, Z. Yin, L. Bai, Reso: A reward-driven self-organizing llm-based multi-agent system for reasoning tasks, in: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, Association for Computational Linguistics, 2025, pp. 15979–15998. doi:10.18653/v1/2025.emnlp-main.808.
 URL <https://aclanthology.org/2025.emnlp-main.808>
- [21] D. Zhang, Z.-Z. Li, M.-L. Zhang, J. Zhang, Z. Liu, Y. Yao, H. Xu, J. Zheng, X. Chen, Y. Zhang, F. Yin, J. Dong, Z. Guo, L. Song, C.-L. Liu, From system 1 to system 2: a survey of reasoning large language models, IEEE Transactions on Pattern Analysis and Machine Intelligence (2025) 1–20doi:10.1109/tpami.2025.3637037.
 URL <https://strathprints.strath.ac.uk/94892/7/Zhang-et-al-IEEE-TPAMI-2025-From-system-1-to-system-2.pdf>
- [22] P. Pitre, G. Srivastava, L. Zhang, L. Wang, N. Ramakrishnan, X. Wang, A diagnostic study of multi-agent llms for real-world debates, in: Proceedings of the 43rd International Conference on Machine Learning, Vol. 306 of PMLR, Seoul, South Korea, 2026.
 URL <https://openreview.net/pdf?id=78Q4xkcJHc>
- [23] K. Ueda, W. Hirota, T. Asakura, T. Omi, K. Takahashi, K. Arima, T. Ishigaki, Exploring the design of multi-agent LLM dialogues for research ideation, in:

- Proceedings of the 26th Annual Meeting of the Special Interest Group on Discourse and Dialogue, Association for Computational Linguistics, Avignon, France, 2025, pp. 322–337.
URL <https://aclanthology.org/2025.sigdial-1.26/>
- [24] Y. Wang, K. Zhang, K. Tian, S. Zeng, X. Lv, N. Ding, B. Qi, B. Zhou, Scalability of LLM-based multi-agent systems for scientific code generation: A preliminary study, in: Proceedings of The 3rd Workshop on Mathematical Natural Language Processing (MathNLP 2025), Association for Computational Linguistics, Suzhou, China, 2025, pp. 50–61. doi:10.18653/v1/2025.mathnlp-main.4.
URL <https://aclanthology.org/2025.mathnlp-main.4/>
- [25] X. Li, S. Wang, S. Zeng, Y. Wu, Y. Yang, A survey on LLM-based multi-agent systems: workflow, infrastructure, and challenges, *Vicinagearth* 1 (9) (2024). doi:10.1007/s44336-024-00009-2.
URL <https://link.springer.com/article/10.1007/s44336-024-00009-2>
- [26] D. Han, K. Doya, D. Li, J. Tani, Synergizing habits and goals with variational bayes, *Nature Communications* 15 (1) (2024) 4461. doi:10.1038/s41467-024-48577-7.
- [27] B. Yao, C. Shang, W. Du, J. He, R. Lian, Y. Zhang, H. Su, S. Swamy, Y. Qi, Peacemaker or troublemaker: How sycophancy shapes multi-agent debate, *arXiv preprint arXiv:2509.23055* (2025).
URL <https://arxiv.org/abs/2509.23055>
- [28] A. Wynn, H. Satija, G. K. Hadfield, Talk isn’t always cheap: Understanding failure modes in multi-agent debate, *arXiv preprint arXiv:2509.05396* (2025).
URL <https://arxiv.org/abs/2509.05396>
- [29] J. C.-Y. Chen, S. Saha, M. Bansal, ReConcile: Round-table conference improves reasoning via consensus among diverse LLMs, in: L.-W. Ku, A. Martins, V. Srikumar (Eds.), Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), Association for Computational Linguistics, Bangkok, Thailand, 2024, pp. 7066–7085. doi:10.18653/v1/2024.acl-long.381.
URL <https://aclanthology.org/2024.acl-long.381/>
- [30] M. Sharma, M. Tong, T. Korbak, D. Duvenaud, A. Askell, S. R. Bowman, et al., Towards understanding sycophancy in language models, in: The Twelfth International Conference on Learning Representations, 2024, arXiv:2310.13548.
URL <https://openreview.net/forum?id=tvhaxkMKAn>

- [31] H. Wang, X. Du, W. Yu, Q. Chen, K. Zhu, Z. Chu, L. Yan, Y. Guan, Learning to break: Knowledge-enhanced reasoning in multi-agent debate system, *Neurocomputing* 618 (2025) 129063. doi:10.1016/j.neucom.2024.129063. URL <https://www.sciencedirect.com/science/article/pii/S0925231224018344>
- [32] Y. Liu, J. Cao, Z. Li, R. He, T. Tan, Breaking mental set to improve reasoning through diverse multi-agent debate, in: Y. Yue, A. Garg, N. Peng, F. Sha, R. Yu (Eds.), *International Conference on Learning Representations*, Vol. 2025, 2025, pp. 24666–24725. URL https://proceedings.iclr.cc/paper_files/paper/2025/file/3de667dab3b3d812583abc0a786139a0-Paper-Conference.pdf
- [33] Y. Li, Y. Du, J. Zhang, L. Hou, P. Grabowski, Y. Li, E. Ie, Improving multi-agent debate with sparse communication topology, in: *Findings of the Association for Computational Linguistics: EMNLP 2024*, Association for Computational Linguistics, Miami, Florida, USA, 2024, pp. 7281–7294. doi:10.18653/v1/2024.findings-emnlp.427. URL <https://aclanthology.org/2024.findings-emnlp.427/>
- [34] S. Gómez Álvarez, A. Mozo Quesada, T. Navarro, S. Gálvez Rojas, F. López Valverde, Multi-agent debate system based on large language models: structured deliberation and validation in satellite communications, *Journal of Intelligent Information Systems* (Aug. 2026). doi:10.1007/s10844-026-01086-z. URL <https://doi.org/10.1007/s10844-026-01086-z>
- [35] Y. Zhou, Y. Chen, Adaptive heterogeneous multi-agent debate for enhanced educational and factual reasoning in large language models, *Journal of King Saud University - Computer and Information Sciences* 37 (2025) 330. doi:10.1007/s44443-025-00353-3.
- [36] W. Yang, L. Some, M. Bain, B. Kang, A comprehensive survey on integrating large language models with knowledge-based methods, *Knowledge-Based Systems* (2025). doi:10.1016/j.knosys.2025.113503.